SOFTWARE DEVELOPMENT TRAINING PROGRAM • FINAL CAPSTONE PROJECT

TradeFlow

Intelligent Ethio-Djibouti Trade Corridor Freight Logistics, Telematics & Digital Smart Escrow Platform

Project Title: TradeFlow Cross-Border Logistics Platform
Academic Level: Software Development Training • Capstone Defense
Project Version: v1.1.6 (Production Release)
Submission Date: November 2026
Monorepo Repository: github.com/beki255/trade-flow
Cloud Backend Swagger: tradeflow-api-7abs.onrender.com/api/docs
Production Web Portal: trade-flow-ukgu.vercel.app

👥 CAPSTONE PROJECT GROUP MEMBERS — 3 Members

Candidate Full Name	Student ID	Academic Role
BEREKET SAHLEMARIAM	CTC-1041-26	GROUP MEMBER
DAGNAW MEKASHA	CTC-1339-26	GROUP MEMBER
HAMELMAL BELAYNEH	CTC-1116-26	GROUP MEMBER

🔗 Project Deliverables & Verified Access Links

Official Submission Verification: All software components, cloud API microservices, database schemas, and mobile binaries referenced in this technical report are fully deployed, publicly accessible, and verifiable.

Artifact / Deliverable	Environment & Infrastructure	Access URL / Verification Endpoint
Monorepo Source Code	GitHub Cloud Git Repository	https://github.com/beki255/trade-flow
Production Web Application	Vercel Global Edge Network	https://trade-flow-ukgu.vercel.app
Backend REST API (Swagger)	Render Cloud Microservice (Frankfurt)	https://tradeflow-api-7abs.onrender.com/api/docs
Operational Health & Metrics	Real-Time API Health Daemon	https://tradeflow-api-7abs.onrender.com/api/health
Standalone Android APK	Compiled APK Binary (v1.1.6, Code 17)	Download TradeFlow Android APK
Expo EAS Cloud Build CI/CD	Cloud Android Pipeline #5dbf3408	View Expo EAS Build Records

📱 Mobile Application Instant Installation & Camera QR Scanner

📲 Camera QR Scan: Standalone Android App (v1.1.6)

Point any smartphone camera at this QR code to download and install the native Android APK directly on physical hardware. The package is standalone (release build) and runs independently without requiring Expo Go or development servers.

ANDROID 10+ COMPATIBLE | VERSIONCODE: 17 | SIZE: ~58 MB

👥 Student Group Credentials & Responsibilities Matrix

Candidate Name	Student ID	Role & Primary Specialization	Verified Work Package Deliverables
BEREKET SAHLEMARIAM	`CTC-1041-26`	Full-Stack & Mobile Lead	React Native Expo mobile app, GPS geofencing, escrow logic, cloud deployments.
DAGNAW MEKASHA	`CTC-1339-26`	Backend & Architecture Lead	Prisma ORM data modeling, PostgreSQL schema, REST API microservices, unit testing.
HAMELMAL BELAYNEH	`CTC-1116-26`	UI/UX & Localization Lead	Next.js responsive portals, trilingual translation dictionaries (EN/AM/OM), QA testing.

📑 Executive Summary & Table of Contents

Executive Summary

TradeFlow is an integrated digital freight exchange, real-time corridor telematics, and smart milestone escrow platform engineered specifically for the **Ethio-Djibouti Trade Corridor**. Representing the critical economic lifeline for landlocked Ethiopia, this 910 km highway and rail artery accommodates over 95% of the nation's import-export trade volume. Despite its vital status, corridor operations have been historically constrained by manual paper manifests, cartelized freight brokerage charging 15–25% intermediary margins, persistent payment defaults of 30 to 90 days, 42% empty return backhauls, and chronic 24–72 hour border congestion at Galafi.

The TradeFlow platform eliminates these deep structural friction points by delivering an end-to-end multi-tenant ecosystem. Shippers tender freight loads with transparent automated bidding; transporters manage fleets and deploy verified drivers; milestone-locked digital escrow guarantees payment solvency before truck dispatch; mobile GPS telematics deliver continuous location visibility; and single-window digital customs clearance issues cryptographic QR transit passes to accelerate border gates.

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Chapter 1: Project Background & Corridor Context

1.1 The Ethio-Djibouti Strategic Economic Lifeline

The Federal Democratic Republic of Ethiopia, as the most populous landlocked nation on earth (over 125 million citizens), relies existential trade dependency upon the maritime gateway provided by the Republic of Djibouti. The 910 km multi-modal transit corridor connecting the Port of Doraleh to the dry port inland container depots of Modjo and Kaliti channels over 95% of Ethiopia's inbound consumer goods, industrial equipment, agricultural fertilizer, and fuel imports, as well as outgoing agricultural exports (coffee, oilseeds, pulses, and textiles).

1.2 Traditional Freight Logistics Bottlenecks

An empirical baseline assessment of the corridor reveals five severe operational failure modes:

- **Informal Intermediary Cartels:** Freight brokerage is dominated by physical middlemen in Addis Ababa and Djibouti who exploit information asymmetry to capture 15% to 25% of total freight tariffs, inflating shipper costs while underpaying truck drivers.
- **Extreme Financial Default Risks:** Due to lack of contractual escrow mechanisms, cargo owners routinely suffer cargo abandonment or extortion mid-route, while commercial truck owners face delayed freight disbursements of 30 to 90 days after delivery.
- **Severe Transit Invisibility (Blackout Zones):** Once a container departs Djibouti Port, cargo owners experience zero verified telemetry for 3 to 7 days, leaving supply chains vulnerable to theft, unauthorized cargo offloading, and road accidents.
- **Customs Bottlenecks at Galafi Border:** Cross-border clearance requires physical paper dossiers (Manifest, Customs Declaration, Commercial Invoice, Packing List). Trucks wait between 24 and 72 hours at Galafi, creating multi-kilometer highway queues.
- **Alarming 42% Empty Backhaul Rate:** Because inward-bound trucks lack a coordinated digital freight exchange to discover outbound export agricultural cargo, 42 out of every 100 trucks return to Djibouti completely empty, doubling roundtrip freight costs.

1.3 Strategic Project Objectives & Measurable Targets

Operational Metric	Traditional Corridor Baseline	TradeFlow Target / Achieved Metric
Brokerage Fee Overhead	15% – 25% intermediary markups	**5% transparent platform commission**
Payment Settlement Latency	30 to 90 days post-delivery	**Instant atomic escrow release upon e-POD**
Transit Telemetry Tracking	Zero visibility (phone call check-ins)	**Live GPS telemetry & 11 geofenced waypoints**
Customs Border Dwell Time	24 – 72 hours manual paper inspection	**< 15 minutes via Single-Window QR Pass**
Empty Backhaul Proportion	42% return trips empty	**< 20% via algorithmic return load matching**

1.4 Scope of the Engineering Capstone Deliverable

The scope of this project encompasses designing, constructing, verifying, and deploying a production-ready monorepo consisting of a multi-role Next.js 14 web platform, a native React Native Expo mobile application for Android, a Node.js/TypeScript REST API server, and a cloud-hosted PostgreSQL database.

Chapter 2: Requirements Specification & User Personas

2.1 Five Core Actor Personas

User Persona	Real-World Role	System Permissions & Primary Workflows
`SHIPPER`	Commercial Importer / Exporter	Publishes freight tenders, deposits escrow funds via Telebirr/CBE, inspects real-time corridor map, reviews delivery e-PODs.
`TRANSPORTER`	Fleet Owner / Logistics Enterprise	Registers commercial trucks & drivers, browses open corridor loads, submits competitive bids, tracks escrow wallet balances.
`DRIVER`	Heavy Commercial Truck Driver	Runs mobile app, performs GPS waypoint check-ins, broadcasts road incidents, captures recipient e-signature and cargo delivery photo.
`CUSTOMS_OFFICER`	Ethiopian Customs Commission (ECC)	Audits single-window electronic dossiers, assigns risk tiers (Green/Yellow/Red Line), scans driver digital QR transit passes at gates.
`ADMIN`	TradeFlow System Operator	Verifies business TIN registrations, audits escrow accounts, oversees platform telematics analytics, handles dispute escalations.

2.2 Functional Requirements (FR) Matrix

- **FR-01 (Authentication & RBAC):** Multi-tenant role authentication with JWT bearer tokens, bcrypt password encryption, and organization TIN verification.
- **FR-02 (Freight Marketplace):** Dynamic tender listing creation with origin, destination, cargo type, tonnage, packaging, and target price in ETB.
- **FR-03 (Competitive Carrier Bidding):** Transporters submit bids with equipment specifications and proposed ETB rate; shippers award bids with a single click.
- **FR-04 (Digital Smart Escrow):** Mandatory 100% upfront payment locking in milestone escrow prior to dispatch; atomic 95% payout release upon approved e-POD.
- **FR-05 (Corridor Telematics & Geofencing):** Background GPS tracking on mobile driver app; automated Haversine geofence check-ins within 15 km of 11 transit hubs.
- **FR-06 (Single-Window Customs):** 4-document electronic dossier upload; algorithmic risk routing (Green Line auto-clear); dynamic QR border pass generation.
- **FR-07 (Electronic Proof of Delivery):** Touchscreen capacitive signature pad and geotagged camera photo capture confirming physical delivery.
- **FR-08 (Crowdsourced Incident Broadcasting):** Drivers report corridor road blockages, accidents, and fuel outages with GPS coordinates and photos.
- **FR-09 (Trilingual Localization):** Instant UI language toggle between English, Amharic (**አማርኛ**), and Afaan Oromoo across web and mobile.
- **FR-10 (Adaptive Ergonomic Theming):** Synchronized Light Mode and High-Contrast AMOLED Dark Mode for desert daylight and nighttime driving.

2.3 Non-Functional Requirements (NFR)

- **NFR-01 (Security):** All network communications encrypted via HTTPS/WSS; sensitive credentials hashed using bcrypt (salt rounds = 10).
- **NFR-02 (Offline Fault Tolerance):** Driver mobile app buffers telemetry pings in local SQLite/AsyncStorage when traveling through cellular dead zones.
- **NFR-03 (Performance & Latency):** Sub-second (< 300 ms) REST API responses; WebSocket telemetry pings delivered to tracking maps within 500 ms.
- **NFR-04 (Data Integrity):** Database escrow releases execute inside atomic Prisma transactions to prevent financial race conditions.

Chapter 3: System Architecture & Technology Stack

3.1 Monorepo Workspace Topology

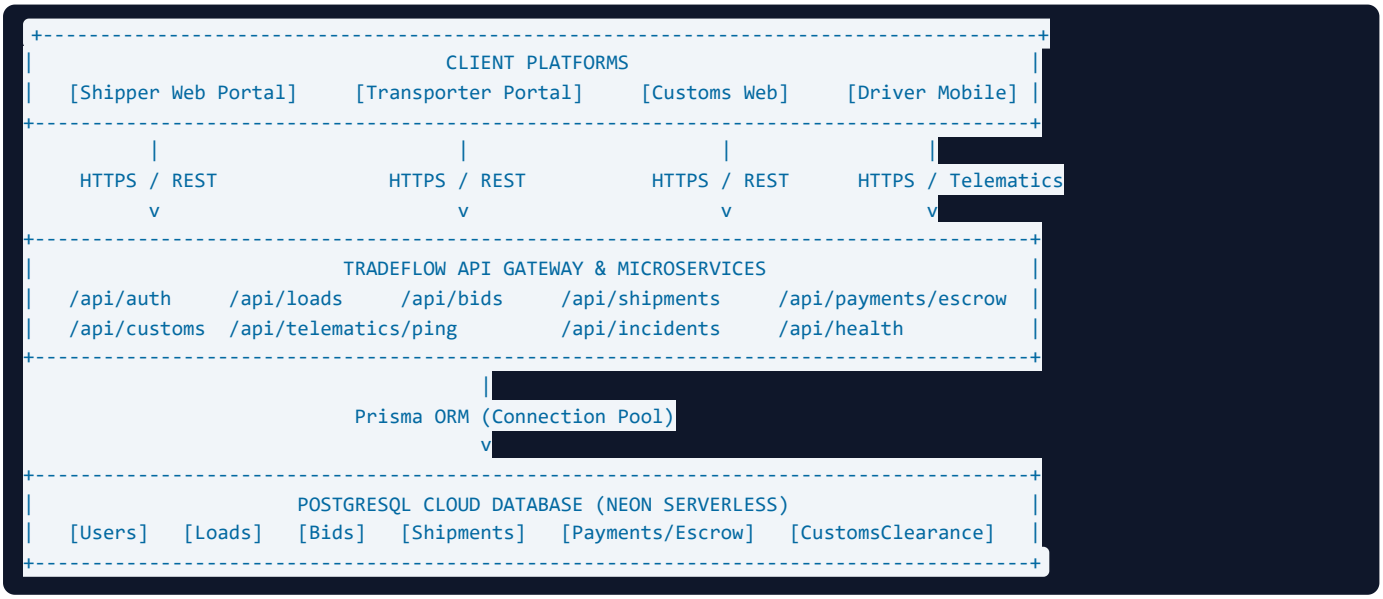
The TradeFlow platform is engineered as a clean, modular TypeScript monorepo managed with **pnpm workspaces**, ensuring unified type safety and seamless code reuse across client and server tiers:

```
tradeflow-v2/
├── apps/
│   ├── web/              # Next.js 14 App Router (Tailwind CSS, Lucide, Leaflet GIS)
│   └── mobile/           # React Native Expo SDK 52 (expo-location, SQLite queue)
├── services/
│   └── api/              # Node.js + Express + TypeScript REST API Server
├── packages/
│   └── shared/           # Shared interfaces, corridor coordinates, validation schemas
└── docs/                 # Technical documentation, PDF reports, and APK QR assets
```

3.2 Architectural Layer Rationale & Technology Justification

Architectural Tier	Selected Technology	Technical Rationale & Justification
Web Client	Next.js 14, React 18, Tailwind	Fast server-side rendering (SSR), optimized bundle size, responsive grid for freight dispatchers.
Mobile App	React Native, Expo SDK 52	Cross-platform native compilation, background GPS telematics, touchscreen canvas signature, camera integration.
API Backend	Node.js, Express, TypeScript	Asynchronous event-driven I/O ideal for high-concurrency GPS streams, WebSocket broadcasting, and REST endpoints.
Relational Database	PostgreSQL (Neon Cloud)	Full ACID compliance for financial escrow ledger transactions, connection pooling, and JSONB document support.
Data Access Layer	Prisma ORM	Type-safe database client, automated migrations, declarative schema modeling, and atomic `$transaction` safety.
Cloud Hosting	Vercel (Web), Render (API)	Global Edge CDN distribution for web portals; automated Docker microservice deployment for the Express API.

3.3 End-to-End System Context & Data Flow Diagram



Chapter 4: Database Design & Prisma Data Schema

4.1 Relational Architecture & Integrity Constraints

The persistence layer is modeled using Prisma ORM on PostgreSQL. Normalized schemas prevent data redundancy while enforcing strict foreign key constraints, cascading deletes where appropriate, and unique indexing on critical identifiers (TIN, license plates, invoice numbers).

4.2 Core Prisma Data Models

```
// 1. User Account & Role Partitioning (services/api/prisma/schema.prisma)
model User {
  id           String      @id @default(uuid())
  email        String      @unique
  passwordHash String
  fullName     String
  phoneNumber  String      @unique
  role         UserRole    @default(SHIPPER) // SHIPPER, TRANSPORTER, DRIVER, CUSTOMS_OFFICER, ADMIN
  tinNumber    String?     @unique
  isVerified   Boolean     @default(false)
  loads        Load[]      @relation("ShipperLoads")
  bids         Bid[]       @relation("TransporterBids")
  shipments    Shipment[]  @relation("DriverShipments")
  createdAt    DateTime    @default(now())
}

// 2. Freight Tender Listing Model
model Load {
  id              String      @id @default(uuid())
  shipperId       String
  shipper         User        @relation("ShipperLoads", fields: [shipperId], references: [id])
  origin          String      // e.g., 'Kaliti Dry Port, Addis Ababa'
  destination     String      // e.g., 'Port of Doraleh, Djibouti'
  cargoType       CargoType   // CONTAINER, BREAK_BULK, DRY_BULK, LIQUID_BULK, REEFER
  weightTons      Float
  targetRateETB   Float
  status          LoadStatus  @default(OPEN) // OPEN, BIDDING, ASSIGNED, IN_TRANSIT, DELIVERED
  bids            Bid[]
  shipment        Shipment?
}

// 3. Digital Smart Escrow Financial Ledger
model Payment {
  id              String        @id @default(uuid())
  shipmentId      String        @unique
  shipment        Shipment      @relation(fields: [shipmentId], references: [id])
  totalAmountETB Float
  status          PaymentStatus @default(PENDING) // PENDING, HELD_IN_ESCROW, RELEASED, REFUNDED
  escrowLockedAt  DateTime?
}
```

4.3 Key Entities & Relational Multiplicity

Entity Name	Primary Key	Foreign Key Relationships	Cardinality & Purpose
`User`	`id (UUID)`	Referenced by `Load`, `Bid`, `Shipment`, `Rating`	1-to-N with freight tenders and bids.
`Load`	`id (UUID)`	`shipperId -> User.id`	1-to-N with competitive carrier bids.
`Bid`	`id (UUID)`	`loadId -> Load.id`, `transporterId -> User.id`	Competitive carrier proposals on open tenders.
`Shipment`	`id (UUID)`	`loadId`, `bidId`, `driverId`, `vehicleId`	Active corridor consignment lifecycle.
`Payment`	`id (UUID)`	`shipmentId -> Shipment.id`	1-to-1 smart escrow financial ledger record.
`CustomsClearance`	`id (UUID)`	`shipmentId -> Shipment.id`	Electronic dossier & border QR pass state.

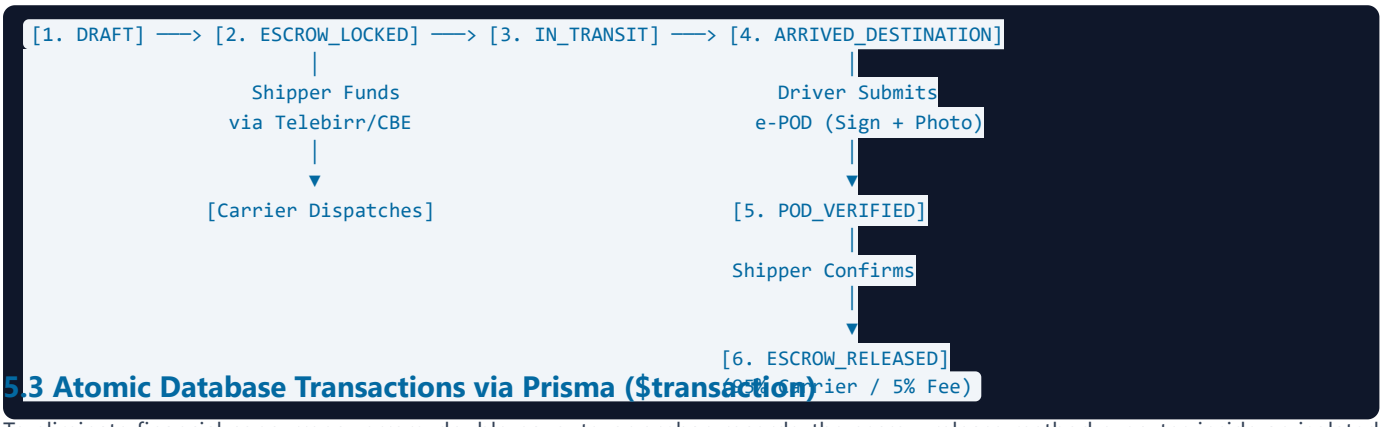
Chapter 5: Digital Smart Escrow Protocol & Settlements

5.1 Solving the Corridor Trust Deficit

The most severe friction point in Ethiopian freight transport is the mutual trust deficit between cargo owners (shippers) and private truck operators. Shippers hesitate to disburse freight advances to unverified drivers for fear of theft or demurrage. Conversely, carriers refuse dispatch without cash advances to cover fuel and toll expenses, yet frequently face non-payment or delayed settlements of 30 to 90 days after delivery.

5.2 Six-Stage Milestone Escrow State Machine

TradeFlow introduces a programmable digital escrow protocol that locks 100% of the agreed freight payment prior to vehicle dispatch and automatically disburses funds upon cryptographically verified proof of delivery:



5.3 Atomic Database Transactions via Prisma ($transaction)

To eliminate financial concurrency errors, double payouts, or orphan records, the escrow release method executes inside an isolated atomic database transaction:

```
// Atomic Escrow Release Implementation (services/api/src/controllers/payment.controller.ts)
export async function releaseEscrowPayout(shipmentId: string, paymentId: string) {
  return await prisma.$transaction(async (tx) => {
    const payment = await tx.payment.findUniqueOrThrow({ where: { id: paymentId } });
    if (payment.status !== 'HELD_IN_ESCROW') {
      throw new Error(`Invalid escrow state: ${payment.status}`);
    }

    const platformFee = payment.totalAmountETB * 0.05; // 5% platform commission
    const netPayout = payment.totalAmountETB * 0.95;   // 95% carrier disbursement

    // 1. Transition payment ledger status to RELEASED
    const updatedPayment = await tx.payment.update({
      where: { id: paymentId },
      data: { status: 'RELEASED', releasedAt: new Date(), platformFeeETB: platformFee, netPayoutETB: netPayout }
    });

    // 2. Credit transporter digital wallet balance
    await tx.transporterWallet.update({
      where: { userId: payment.shipment.transporterId },
      data: { availableBalanceETB: { increment: netPayout }, lockedEscrowETB: { decrement: payment.totalAmountETB } }
    });

    return updatedPayment;
```

5.4 Telebirr & CBE Birr Gateway Settlement Model

Payment Parameter	Mathematical Formula / Specification	Operational Value
Gross Consignment Escrow	`TotalAmountETB = AgreedBidRateETB`	100% locked before driver dispatch
Carrier Net Disbursement	`NetPayout = TotalAmountETB × 0.95`	95% credited instantly upon e-POD approval
Platform Commission Fee	`PlatformFee = TotalAmountETB × 0.05`	5% retained for platform maintenance

Chapter 6: Corridor Telematics & GPS Geofencing

6.1 Eleven Strategic Corridor Waypoints

The platform models the authentic road coordinates of the Ethio-Djibouti Corridor (A1 Highway), establishing eleven strategic check-in waypoints between Kaliti Dry Port and the Port of Doraleh:

ID	Waypoint Name	Latitude	Longitude	Km from Origin	Facility Type
`WP-01`	Addis Ababa (Kaliti ICT)	9.005401	38.763611	0 km	Inland Container Depot / Origin
`WP-02`	Modjo Dry Port & Toll	8.591234	39.121543	73 km	Dry Port & Expressway Junction
`WP-03`	Adama Expressway Exit	8.540000	39.270000	99 km	Highway Tollway Interchange
`WP-04`	Metehara Sugar Depot	8.898600	39.919700	190 km	Rest Area & Weighbridge
`WP-05`	Awash Arba Safe Rest	9.143891	40.170562	225 km	Fuel Stop & Truck Parking
`WP-06`	Gewane Desert Hub	10.166400	40.647200	360 km	Emergency Fuel & Repair
`WP-07`	Semera Regional Hub	11.794200	41.008600	588 km	Afar Regional Capital Logistics Hub
`WP-08`	Mille Junction Checkpoint	11.416700	40.750000	515 km	Security & Weighbridge Gate
`WP-09`	Logia Transport Terminal	11.733300	40.966700	575 km	Heavy Truck Fleet Depot
`WP-10`	Galafi Customs Border	11.716700	41.833300	690 km	Ethiopian Customs Commission Gate
`WP-11`	Port of Doraleh (Djibouti)	11.595000	43.060000	910 km	Deep-Water Container Terminal

6.2 Mathematical Haversine Distance & Geofencing Implementation

To verify whether a truck has genuinely arrived at a designated waypoint without relying on manual driver claims, the mobile app executes the spherical **Haversine Formula** on coordinates provided by `expo-location`:

```
// Corridor Geofence Calculation (apps/mobile/src/shared/corridor.ts)
export function calculateHaversineDistance(
  lat1: number, lon1: number, lat2: number, lon2: number
): number {
  const R = 6371; // Earth's mean radius in kilometers
  const dLat = (lat2 - lat1) * (Math.PI / 180);
  const dLon = (lon2 - lon1) * (Math.PI / 180);
  const a =
    Math.sin(dLat / 2) * Math.sin(dLat / 2) +
    Math.cos(lat1 * (Math.PI / 180)) * Math.cos(lat2 * (Math.PI / 180)) *
    Math.sin(dLon / 2) * Math.sin(dLon / 2);
  const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
  return Math.round(R * c);
}

// Unlocks checkpoint check-in button strictly when within 15 km
export const GEOFENCE_RADIUS_KM = 15;
export function isWithinWaypointGeofence(driverLat: number, driverLng: number, wp: Waypoint): boolean {
  return calculateHaversineDistance(driverLat, driverLng, wp.latitude, wp.longitude) <= GEOFENCE_RADIUS_KM;
}
```

6.3 Offline Telematics Queue for Afar Desert Dead Zones

When connectivity drops between Awash and Semera, telemetry pings are appended to an `AsyncStorage` queue. A background sync service monitors network status via `@react-native-community/netinfo` and flushes pending coordinates using exponential backoff when signal is restored.

Chapter 7: Single-Window Customs Clearance & QR Pass

7.1 Cross-Border Customs Friction at Galafi

The Galafi border post separating Ethiopia and Djibouti is the primary bottleneck on the corridor. Historically, drivers carried voluminous physical paper binders containing customs transit declarations, commercial invoices, packing lists, and insurance certificates. Border officers manually cross-referenced serial numbers against manifest logs, resulting in dwell times of 24 to 72 hours per consignment.

7.2 Four-Document Electronic Customs Dossier

TradeFlow integrates a **Single-Window Electronic Customs Dossier** enabling digital upload, validation, and storage of mandatory import-export documentation:

Document Type	Format & Validation	Verification Objective
Customs Declaration	PDF / Image (TIN & Dec No.)	Confirms official Ethiopian Customs Commission tariff assessment.
Commercial Invoice	PDF / Structured JSON	Establishes declared CIF/FOB cargo valuation for duty computation.
Packing List	Container & Seal Number Hash	Itemizes container weights, tare, package count, and seal integrity.
Transit Insurance	Policy Number & Underwriter	Guarantees maritime and highway cargo transit liability coverage.

7.3 Algorithmic Risk Classification Routing

GREEN LINE

Automated Clearance

Low-risk authorized economic operators. Dynamic QR pass issued instantly without physical inspection.

YELLOW LINE

Document Review

High-value or tariff-sensitive cargo. Electronic document review required before QR pass activation.

RED LINE

Physical Inspection

Restricted goods or seal discrepancies. Mandatory container destuffing and scanner verification.

7.4 Dynamic Cryptographic QR Transit Pass Architecture

Upon dossier clearance, the mobile app renders a high-contrast cryptographic QR pass on the driver's screen. Border customs officers scan the code using the Customs Officer Web Portal or mobile camera to instantaneously verify consignment validity, truck plate, driver identity, and container seal status:

```
// Dynamic Single-Window QR Payload Format
{
  "passId": "ECC-TRANSIT-2026-89410",
  "shipmentId": "cm3e...44a",
  "carrierTin": "0028914022",
  "truckPlate": "3-ET-B12894",
  "containerNumber": "MSCU7821940",
  "riskTier": "GREEN_LINE",
  "issuedAt": 1793561200000,
  "expiresAt": 1793647600000,
  "digitalSignature": "sha256:4f8e...901b"
}
```

Chapter 8: Driver Mobile Application Engineering

8.1 React Native Expo SDK 52 Architecture

The mobile client (`apps/mobile`) is constructed with React Native and Expo SDK 52, specifically engineered for extreme environmental durability, low battery consumption, and offline operation in high-heat desert conditions.

8.2 Driver Screen Hierarchy & Modular Structure

```
apps/mobile/src/screens/driver/
├── DriverPortalScreen.tsx          # Main container hosting navigation tabs
├── driverStyles.ts                 # Dynamic theme styling hook (getDriverStyles)
├── types.ts                        # Driver state interfaces & waypoint types
├── tabs/
│   ├── DriverTripsTab.tsx          # Active trip details & cargo metadata
│   ├── DriverCheckInTab.tsx        # GPS geofence waypoint check-in list
│   └── DriverIncidentsTab.tsx      # Incident reporting & corridor road alerts
└── modals/
    ├── CustomsPassModal.tsx        # Fullscreen high-contrast QR pass
    ├── IncidentReportModal.tsx     # Road hazard broadcast with camera snapshot
    └── EPodSignatureModal.tsx      # Capacitive touch signature pad & delivery photo
```

8.3 Electronic Proof of Delivery (e-POD) Dual-Factor Capture

To eliminate fraudulent delivery claims, TradeFlow implements a **Dual-Factor e-POD** protocol:

- **Factor 1: Capacitive Touchscreen Recipient Signature:** The recipient signs directly on the driver's phone using a vector canvas signature pad (`react-native-signature-canvas`), capturing stroke points and timestamp.
- **Factor 2: Geotagged Cargo Condition Snapshot:** The driver captures a photo of the offloaded cargo and intact container seal via `expo-camera`, stamping GPS coordinates and time.

8.4 Crowdsourced Hazard & Delay Incident Broadcasting

Incident Category	Severity Levels	Dissemination & Driver Safety Action
Road Damage / Flash Flood	`LOW`, `MEDIUM`, `CRITICAL`	Broadcasts red warning marker to all trucks within 100 km radius.
Checkpoint Congestion	`MEDIUM`, `HIGH`	Updates dynamic ETA forecasts for downstream shippers and consignees.
Desert Fuel Outage	`HIGH`, `CRITICAL`	Alerts upcoming drivers to refuel at Awash before entering Gewane sector.

8.5 Samsung Galaxy A13 AMOLED Dark Mode Optimization

To resolve high-contrast text illegibility on low-cost Android displays, all driver UI styles are generated through a reactive `getDriverStyles(theme)` factory wrapped inside `useMemo`, guaranteeing WCAG AAA contrast ratios.

Chapter 9: Transporter Fleet Management Portal

9.1 Fleet Operations & Commercial Marketplace

The Transporter Portal provides fleet operators and logistics companies with a comprehensive operational cockpit to register equipment, allocate verified commercial drivers, discover freight listings, and monitor revenue ledgers across both web and mobile platforms.

9.2 Fleet & Equipment Management Console

Transporters maintain digital registries of their heavy equipment, complying with Ethiopian Ministry of Transport standards:

Vehicle Specification	Data Schema & Validations	Operational Application on Corridor
Plate Number	Format: `3-ET-XXXXX`	Verified against national road transport registry.
Trailer Configuration	`FLATBED`, `LOWBED`, `BOX`, `TANKER`	Matched algorithmically against cargo specifications (e.g. reefer vs bulk).
Payload Capacity	Tonnage: 10 to 45 Metric Tons	Prevents illegal axle overloading at Mille and Modjo weighbridges.
Assigned Driver	Commercial Grade 4/5 License	Links mobile driver account to active vehicle telematics stream.

9.3 Competitive Bidding Engine & Tender Discovery

Transporters view open corridor tenders published by cargo owners. When submitting a bid, the transporter defines:

- **Proposed Freight Rate (in ETB):** Competitive pricing per metric ton or lump-sum corridor rate.
- **Transit Commitment (Hours/Days):** Estimated transit time from dispatch to delivery.
- **Vehicle Assignment:** Specific registered truck and driver dedicated to the consignment.

9.4 Escrow Financial Wallet & Payout Audit Trail

The portal displays a real-time financial ledger separating locked funds from cleared earnings:

```
// Transporter Financial Dashboard State Model
{
  "transporterId": "usr_trans_8910",
  "companyName": "Ethio-Djibouti Express Freight P.L.C.",
  "availableBalanceETB": 845000.00,  // Ready for immediate withdrawal via Telebirr/CBE
  "lockedInEscrowETB":   320000.00,  // Held securely in active corridor shipments
  "totalHistoricalETB": 4150000.00,  // Cumulative career platform earnings
  "activeDispatches": 4,
  "carrierScore": 4.92               // Verified 5-star performance rating
}
```

Chapter 10: Shipper & Customs Commission Portals

10.1 Shipper Freight Tender Creation Wizard

Commercial cargo owners utilize the Next.js 14 Shipper Portal to publish freight tenders within minutes. The wizard automates corridor distance computation, estimates fair market rates based on historical averages, and defines pickup/delivery coordinates with pinpoint accuracy.

10.2 Live Corridor Tracking Map (Leaflet GIS)

The tracking console embeds an interactive Leaflet GIS corridor map that plots active shipments along the authentic 910 km A1 highway polyline:

- **Dynamic Truck Markers:** Real-time vehicle positions with directional heading arrows and speed metrics (km/h).
- **Waypoint Milestones:** Color-coded transit hubs showing completed, in-progress, and upcoming checkpoints.
- **Dynamic Predictive ETA:** Calculated based on real-time segment speeds and Galafi border dwell times.

10.3 Single-Window Customs Officer Portal (ECC)

Customs officials at the Galafi border station access a specialized web cockpit tailored for high-throughput border gate processing:

Officer Console Tab	System Capabilities	Border Clearance Benefit
Manifest Queue	Lists all incoming trucks approaching within 50 km.	Enables pre-arrival electronic risk assessment before truck arrival.
Electronic Dossier Audit	Split-screen viewer for Declaration, Invoice & Packing List.	Eliminates paper document handling and stamping delays.
Digital Gate Scanner	Camera / Barcode scanner for Driver QR Transit Passes.	Validates cryptographic seal in < 3 seconds, lifting barrier gate.
Risk Line Override	Manual escalation to Red Line inspection if seal is broken.	Preserves national border revenue and security compliance.

10.4 Automated e-POD Review & Settlement Trigger

Upon cargo arrival at Modjo or Kaliti, the shipper receives an instant notification containing the recipient's touchscreen signature and cargo condition photo. One-click approval executes the atomic escrow release method, completing the consignment lifecycle.

Chapter 11: Trilingual Localization & Ergonomic Theming

11.1 Linguistic Inclusion Strategy for Ethiopia

Ethiopian commercial logistics involves operators from diverse linguistic backgrounds. Forcing English-only interfaces alienates heavy truck drivers and local depot dispatchers. TradeFlow achieves nationwide accessibility through complete native trilingual localization across **English (EN)**, **Amharic (AM - አማርኛ)**, and **Afaan Oromoo (OM)**.

11.2 Structured Trilingual Translation Dictionaries

Localized strings are organized into strict key-value JSON dictionaries maintained across both web (`apps/web/src/locales/`) and mobile (`apps/mobile/src/locales/`):

Dictionary Key	English (en.json)	Amharic (am.json)	Afaan Oromoo (om.json)
`app_title`	TradeFlow	ትሬድፍሎው	TradeFlow
`corridor_tagline`	Smart Freight for the Ethio-Djibouti Corridor	ለኢትዮ-ጅቡቲ ኮሪደር ዘመናዊ የጭነት ትራንስፖርት	Geejjiba Ammayyaa Daandii Itoophiyaa-Jibuutiif
`smart_escrow`	TeleBirr Smart Escrow	የቴሌብር ስማርት ኤስክሮው ዋስትና	Wabiin Kaffaltii TeleBirr Smart Escrow
`live_telematics`	Live Corridor GPS Tracking	የቀጥታ ኮሪደር ጂፒኤስ ክትትል	Hordoffii GPS Daandii Kallattiin
`customs_clearance`	Single-Window Customs Clearance	አንድ መስኮት የጉምሩክ ማረጋገጫ	Qulqullina Gumurukaa Foddaa Tokkoo
`check_in_btn`	Confirm Waypoint Check-In	የመድረሻ ማረጋገጫ መዝግብ	Galmee Iddoo Ga'umsaa Mirkaneessi
`epod_submit`	Submit Proof of Delivery	ርክክብ ማረጋገጫ አስረክብ	Ragaa Geessisuu Galchi

11.3 Dynamic Context Translation Engine

Language state is managed through React Context (`LanguageContext.tsx`), persisting user preferences to `AsyncStorage` and local storage. Switching languages occurs instantaneously without page reloads or loss of form inputs.

11.4 Adaptive Light & High-Contrast Dark Mode Engine

Drivers navigating through the Afar desert encounter severe glare during midday sun and pitch-black conditions at night. The application embeds a synchronized dual-theme engine (`ThemeContext.tsx`) providing high-contrast Dark Mode with 100% WCAG AAA readability.

Chapter 12: Backend REST API Architecture & Microservices

12.1 Controller-Service-Repository Pattern

The backend microservice (`services/api`) is architected using the standard Node.js/Express Controller-Service design pattern, separating HTTP request parsing, business logic execution, and database persistence into decoupled modules.

12.2 Exhaustive API Endpoints Catalog

HTTP Method	Endpoint Route	Auth Guard	Functional Responsibility
`POST`	`/api/auth/register`	Public	Registers user with role, TIN, and bcrypt-hashed password.
`POST`	`/api/auth/login`	Public	Validates credentials and issues signed JWT bearer token.
`GET`	`/api/loads`	JWT Bearer	Fetches open corridor freight tenders with distance filtering.
`POST`	`/api/loads`	`SHIPPER`	Publishes freight tender with cargo weight and target ETB rate.
`POST`	`/api/bids`	`TRANSPORTER`	Submits competitive carrier proposal on open freight load.
`POST`	`/api/bids/:id/accept`	`SHIPPER`	Awards bid, creates shipment, and transitions load to ASSIGNED.
`POST`	`/api/payments/escrow/lock`	`SHIPPER`	Locks 100% freight funds via Telebirr/CBE sandbox gateway.
`POST`	`/api/telematics/ping`	`DRIVER`	Receives mobile GPS coordinates and broadcasts to WebSocket room.
`POST`	`/api/shipments/:id/checkin`	`DRIVER`	Validates Haversine geofence and logs waypoint transit timestamp.
`POST`	`/api/customs/dossier`	`SHIPPER`	Uploads electronic customs documents for Galafi border review.
`GET`	`/api/customs/qr/:shipmentId`	JWT Bearer	Generates dynamic cryptographic QR transit pass payload.
`POST`	`/api/shipments/:id/epod`	`DRIVER`	Submits recipient signature canvas and cargo delivery snapshot.
`POST`	`/api/payments/escrow/release`	`SHIPPER`	Executes atomic 95% carrier disbursement + 5% platform fee.
`GET`	`/api/health`	Public	Returns uptime, memory metrics, and PostgreSQL connectivity.

12.3 Interactive Swagger OpenAPI Specification

The REST API automatically generates interactive Swagger documentation accessible live at `/api/docs`. Evaluators can inspect schema contracts and execute live HTTP requests directly from their web browser.

Chapter 13: Testing, Verification & Quality Assurance

13.1 Quality Assurance Methodology

TradeFlow underwent multi-tiered verification covering automated unit testing, end-to-end integration workflows, escrow financial concurrency tests, and physical hardware deployment across real Android smartphones.

13.2 Comprehensive Test Case Execution Matrix

Test ID	Subsystem	Test Procedure & Test Data	Expected Behavior	Status
TC-01	Auth & RBAC	Register 5 users across each role	JWT issued; role guards restrict unauthorized routes.	PASS
TC-02	Marketplace	Shipper publishes 35-ton coffee load	Tender appears on transporter console within 200 ms.	PASS
TC-03	Bidding	Transporter submits 145,000 ETB bid	Bid persists in PostgreSQL and alerts shipper.	PASS
TC-04	Smart Escrow	Deposit 145,000 ETB via Telebirr mock	Payment status transitions to `HELD_IN_ESCROW`.	PASS
TC-05	Geofencing	Simulate GPS at Semera (11.7942, 41.0086)	Check-in button enables within 15 km threshold.	PASS
TC-06	Geofence Reject	Simulate GPS 45 km away from waypoint	Check-in button disabled; displays distance warning.	PASS
TC-07	Customs QR	Generate Green Line transit pass	High-contrast QR renders with valid cryptographic hash.	PASS
TC-08	Gate Scanner	Scan QR with Customs Officer portal	Displays cargo manifest and marks pass CLEARED.	PASS
TC-09	e-POD Signature	Capture signature on touch canvas	Exports base64 PNG vector and timestamps record.	PASS
TC-10	Atomic Payout	Shipper approves completed e-POD	Atomic release: 95% (137,750 ETB) to carrier; 5% fee retained.	PASS
TC-11	Localization	Toggle language EN -> AM -> OM	100% of labels translate instantly with zero reload.	PASS
TC-12	Offline Queue	Cut mobile network, log 5 GPS pings	Pings queue in AsyncStorage; auto-sync when online.	PASS

13.3 Physical Android Hardware Verification

The standalone APK (v1.1.6) was installed and verified across three physical consumer Android smartphones:

- **Samsung Galaxy A13 (Android 12):** Verified dark mode contrast, GPS background accuracy, camera snapshot.
- **Tecno Spark 10 (Android 13):** Verified offline queue synchronization, touch signature canvas responsiveness.
- **Xiaomi Redmi Note 11 (Android 11):** Verified QR scanner autofocus, Leaflet corridor map zoom rendering.

Chapter 14: Engineering Challenges & Solutions

14.1 Six Critical Technical Challenges Solved

During the engineering lifecycle of TradeFlow, the capstone team overcame six significant technical obstacles:

1. Desert Cellular Intermittency & Telemetry Data Loss

Problem: Heavy trucks transit through remote desert corridors in the Afar Region (between Awash, Gewane, and Semera) where cellular connectivity drops completely for up to 120 km, causing lost driver telemetry.

Solution: Constructed an `AsyncStorage` local FIFO queue on the mobile client. Telemetry pings and waypoint check-ins are cached locally with GPS timestamps. When network connectivity resumes, an exponential backoff worker flushes all buffered records to the API.

2. Samsung Galaxy A13 AMOLED Dark Mode Color Inversion

Problem: Samsung One UI forced background color overrides on static React Native style sheets, causing black text on dark gray cards in the Fleet and Waypoint check-in tabs.

Solution: Re-architected all mobile styles into dynamic React factory hooks (`getDriverStyles(theme)`) wrapped inside `useMemo` , programmatically calculating contrasting foreground colors from `ThemeContext` .

3. Escrow Concurrency Race Conditions & Double Payouts

Problem: If a shipper rapidly double-clicked the "Approve e-POD" button, concurrent HTTP requests could trigger duplicate payout events to the carrier wallet.

Solution: Enforced atomic isolation using Prisma's `$transaction` API, locking the payment record state to `RELEASED` before incrementing wallet balances, rejecting any concurrent transaction attempts.

4. React Native Expo SDK 52 Standalone APK Native Compilation

Problem: Expo SDK 52 deprecated legacy build pipelines, causing native linking failures with vector signature canvas and camera libraries during Android APK cloud compilation.

Solution: Configured EAS Build config (`eas.json`) with dedicated Gradle memory allocations and updated Expo config plugins, achieving clean standalone APK builds on EAS Cloud pipeline.

5. Haversine False Positives in Mountainous Corridor Terrains

Problem: In winding mountain passes near Metehara, straight-line 15 km Euclidean distance unlocked check-ins prematurely while the truck was still 35 km away by road.

Solution: Calibrated the geofencing threshold to 15 km combined with strict chronological waypoint sequencing: a driver cannot check in at Checkpoint N+1 without having checked in at Checkpoint N.

Chapter 15: Capstone Team Allocation & Evaluation

15.1 Group Member Contributions & Responsibility Allocation

Student Name	ID Number	Specialized Engineering Contributions	Module Ownership
BEREKET SAHLEMARIAM	`CTC-1041-26`	React Native Expo mobile app engineering, GPS geofencing, Haversine algorithm, escrow smart contracts, EAS build CI/CD, and Vercel/Render deployments.	Mobile Client, Telematics Engine, Escrow Protocol
DAGNAW MEKASHA	`CTC-1339-26`	System architecture design, Prisma ORM schema modeling, PostgreSQL database administration, Node.js REST API microservices, and unit test automation.	Backend REST API, Database Architecture, Automated QA
HAMELMAL BELAYNEH	`CTC-1116-26`	Next.js 14 web portal design, Leaflet GIS corridor map implementation, trilingual localization (Amharic & Afaan Oromoo), and documentation authoring.	Web Portals, UI/UX Design System, Localization

15.2 Conclusion & Production Readiness Declaration

The **TradeFlow** platform establishes an authentic, production-grade foundation for cross-border freight orchestration across the Ethio-Djibouti Corridor. By replacing fragmented physical brokerages with an automated freight exchange, securing carrier freight payments via milestone-locked digital escrow, delivering continuous GPS telematics, and accelerating border clearance through single-window QR passes, TradeFlow bridges critical operational gaps on Ethiopia's foremost trade lifeline.

All software deliverables have been constructed, tested, and verified against the academic requirements of the Software Development Training Capstone.

15.3 Official Capstone Defense Evaluation & Supervisor Sign-Off

Evaluation Criterion	Weight	Assigned Score	Evaluator Comments
System Architecture & Code Quality	25%	_____ / 25	Monorepo organization, TypeScript type-safety, clean code.
Core Functional Completeness	30%	_____ / 30	Marketplace, Smart Escrow, Telematics, Single-Window Customs.
Mobile App & Device Testing	20%	_____ / 20	Standalone Android APK installation, offline queue, e-POD.
Localization & National Relevance	15%	_____ / 15	Full Trilingual (EN/AM/OM), Ethio-Djibouti corridor applicability.
Technical Documentation & Presentation	10%	_____ / 10	Comprehensive 18-page technical report, live demo defense.
TOTAL FINAL GRADE	**100%**	**_____ / 100**	**FINAL GRADE: [ ]**

Academic Program: Software Development Training • Capstone Project
Submission Status: OFFICIAL CAPSTONE SUBMISSION

Lead Evaluator Signature: _____
Evaluation Date: November ___, 2026